Product Design Requirements (PDR)

Project: Bilingual 3D Model Sharing Website with AR & Admin Panel

Languages: Georgian (KA) & English (EN)

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Date: 16 Mar 2026

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1) Executive Summary

Build a modern, mobile-first website for showcasing and sharing 3D models with a built-in 3D/AR viewer, plus two service pages (Reverse Engineering; 3D Scanning) and a Contact page with a form. The site must be fully bilingual (Georgian/English) with a language switch. An Admin Panel (separate login) must enable easy upload and management of 3D models and content, and assignment of models to appear on specific pages.

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2) Goals & Non-Goals

Goals

- Display 3D models in a gallery with a detail view that supports real-time 3D viewing and AR mode (switchable).
- Provide content pages describing Reverse Engineering and 3D Scanning services.
- Provide a Contact page with a form and published phone number (599744774).
- Offer bilingual UI/content (KA/EN) with a language switch.
- Provide an Admin Panel (auth protected) to upload/manage 3D models and decide on which page(s) they appear.

Non-Goals (for v1 unless explicitly added later)

- E-commerce, payments, or quoting engine.
- Public user accounts or community commenting.
- Complex model versioning or collaboration tooling.

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3) Target Users & Use Cases

Users

- **Public visitors** (desktop & mobile) evaluating portfolio and services.
- **Prospects** who want to preview 3D/AR and contact quickly.
- **Admin(s)** uploading models, managing copy, images, and localization.

Key Use Cases

- Browse 3D models → open detail → interact in 3D → switch to AR.
- Read about Reverse Engineering / 3D Scanning services with visual examples.
- Submit a contact form (First name, Last name, Email, Phone) and/or call the displayed number.
- Admin logs in → uploads model files & thumbnails → chooses where to show them (3D Models / Reverse Engineering / 3D Scanning) → publishes.

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4) Information Architecture (IA) & Navigation

Top-level Navigation (both languages)

- **3D Models** (KA: *3D მოდელები*)
- **Reverse Engineering Service** (KA: *რევერსული ინჟინერიის სერვისი*)
- **3D Scanning Service** (KA: *3D სკანირების სერვისი*)
- **Contacts** (KA: *კონტაქტი*)
- Language switch: **EN / KA** (toggle)

Sitemap

- / → Landing (optional hero + quick links to main pages)
- /models → 3D Models gallery
- /models/:slug → Model detail (3D + AR)
- /services/reverse-engineering → Reverse Engineering Service page
- /services/3d-scanning → 3D Scanning Service page
- /contact → Contact page (form + phone number)
- /admin → Admin login

- /admin/* → Admin dashboard & modules

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5) Functional Requirements

5.1 Global

- **Bilingual**: All UI strings and page content maintained in KA and EN.
- **Language switch button** visible in header; persists across pages.
- **Responsive**: Mobile, tablet, desktop breakpoints.
- **SEO**: Localized meta tags, titles, and schema where applicable.

5.2 3D Models Page (/models)

- **Grid gallery** of models with thumbnail, title, short description, and tags.
- **Model cards** clickable → open model detail page.
- (Optional v1.1) Basic filters (tags, categories) and search.

5.3 Model Detail (/models/:slug)

- **Interactive 3D viewer** (orbit, pan, zoom, lighting, fullscreen).
- **AR toggle button** to switch between 3D and AR.
    - Mobile AR opens native viewer where supported (Android Scene Viewer / iOS Quick Look) or WebXR.
    - Desktop: show QR code to open AR on phone.
- **Metadata**: title, description, properties (units/scale), creation date, file size.
- **Media**: optional image gallery alongside the 3D viewer.
- (Optional) **Download** original mesh if allowed (config per model).

5.4 Reverse Engineering Service (/services/reverse-engineering)

- Page content (KA/EN) describing the service.
- **Showcase**: several 3D models visualized as images and/or embedded mini 3D viewers.
- Admin decides which models appear here (from model assignments).

5.5 3D Scanning Service (/services/3d-scanning)

- Page content (KA/EN) describing the service.
- **Gallery**: several pictures (before/after, scans, results).
- Admin can upload and curate the images.

5.6 Contacts (/contact)

- **Form fields** (required unless stated otherwise):
  - First name (KA: *სახელი*)
  - Last name (KA: *გვარი*)
  - Email (KA: *ელ. ფოსტა*)
  - Phone (KA: *ტელეფონი*)
  - **Send** button (KA: *გაგზავნა*)
- **Validation**: email format; basic phone validation; required fields not empty.
- **Submission**: store in database and send notification email to admin inbox.
- **Success/Failure** banners localized.
- **Published phone number** displayed prominently: **599744774** (formatting may add +995 prefix on click-to-call).

5.7 Admin Panel (/admin)

- **Authentication**: username/email + password (see Security).
- **Models Management**:
  - Create/Edit/Delete models.
  - Upload 3D files (GLB/GLTF as primary). Optional USDZ for iOS AR; auto-convert if pipeline is enabled.
  - Upload cover image/thumbnail(s).
  - Fields: title (EN/KA), description (EN/KA), tags, category assignment:
    - Show on: [3D Models], [Reverse Engineering Service], [3D Scanning Service]
  - Toggle: downloadable? visible? featured?
- **Service Pages Content**:

    - WYSIWYG fields per language for both service pages.
    - Upload images for 3D Scanning gallery and RE showcases.
- **Localization**:
    - Manage UI strings (optional) or keep UI strings in code and only manage page content.
- **Contact Submissions**:
    - View list, mark as read, export CSV.
- **Media Library**: manage all uploaded images/files.
- **User Management** (optional v1.1): roles (Admin, Editor).
- **Audit Log** (optional v1.1): track changes.

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6) Content & Localization (EN/KA)

Navigation labels

- EN: 3D Models | Reverse Engineering Service | 3D Scanning Service | Contacts
- KA: 3D მოდელები | რევერსული ინჟინერიის სერვისი | 3D სკანირების სერვისი | კონტაქტი

Buttons

- EN: View in 3D | View in AR | Send | Learn more | Read more
- KA: 3D ნახვა | AR რეჟიმი | გაგზავნა | ვრცლად | წაიკითხე მეტი

Contact form placeholders

- EN: First name | Last name | Email | Phone
- KA: სახელი | გვარი | ელ. ფოსტა | ტელეფონი

System messages

- EN: “Your message has been sent. Thank you!” / “Please fill in all required fields.”
- KA: “თქვენი შეტყობინება გაიგზავნა. გმადლობთ!” / “გთხოვთ, შეავსოთ სავალდებულო ველები.”

SEO: Page titles, meta descriptions, OpenGraph images per language.

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7) Non-Functional Requirements

Performance

- Target LCP < 2.5s on 4G; serve optimized static assets via CDN.
- 3D models: prefer ≤ 10–20 MB per model (hard limit configurable, e.g., 50 MB). Use Draco/KTX2 compression where possible.

Accessibility

- WCAG 2.1 AA: keyboard navigation for viewer controls, alt text for images, sufficient color contrast, focus states.

Compatibility

- Latest evergreen browsers (Chrome, Edge, Firefox, Safari).
- iOS Safari for AR Quick Look (USDZ) or viewer fallbacks.
- Android Chrome for Scene Viewer or WebXR.

Security & Privacy

- HTTPS everywhere; HSTS; secure cookies; CSRF protection on forms.
- Password policy (min length 12, complexity), rate limiting, account lockout after N failures.
- CAPTCHA on contact form (invisible reCAPTCHA or equivalent) optional.
- GDPR alignment: consent for analytics cookies, data retention policy for contact submissions.

SEO & Analytics

- Sitemap.xml, robots.txt, structured data where relevant.
- Google Analytics 4 (or equivalent privacy-aware analytics) with consent banner.

Hosting & DevOps

- Cloud hosting with CDN (Azure/AWS/GCP). Image & model storage on object storage (Azure Blob/AWS S3) with signed URLs.
- CI/CD pipeline with staging and production environments.

- Daily backups for DB and media; retention 30–90 days.

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8) Technology Recommendations (proposed)

Frontend

- **Framework**: Next.js (React) with built-in i18n routing.
- **3D/AR**: <model-viewer> component for GLTF/GLB with AR support (Android Scene Viewer / iOS Quick Look), plus WebXR fallback; or Three.js + custom AR integration for advanced control.
- **Styling**: Tailwind CSS or CSS Modules. Component library optional.

Backend / Admin

- **Headless CMS**: Strapi or Directus for rapid admin panel setup, i18n, and media handling **or** custom Node.js (NestJS/Express) admin if bespoke features are required.
- **Database**: PostgreSQL.
- **Storage**: S3/Azure Blob for large files; CDN in front.
- **Email**: SMTP provider (e.g., SendGrid) for contact notifications.

Rationale: Next.js + model-viewer minimizes AR complexity and supports KA/EN easily. A headless CMS accelerates Admin Panel delivery while maintaining flexibility.

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9) Data Model (Core)

Model

- id
- slug
- title: { en, ka }
- description: { en, ka }
- tags: [string]
- files: {
    - glb_url (required)

    - gltf_url (optional)
    - usdz_url (optional, for iOS AR)

}

- thumbnail_url
- gallery_images: [url]
- show_on: enum[] → [models, reverse_engineering, scanning]
- downloadable: boolean
- visible: boolean, featured: boolean
- created_at, updated_at

Service Pages

- Reverse Engineering: title { en, ka }, body { en, ka }, showcase_models [ModelRef], images [url]
- 3D Scanning: title { en, ka }, body { en, ka }, images [url]

Contact Submission

- id, first_name, last_name, email, phone, created_at, status

Users (Admin)

- id, email, password_hash, role, last_login_at

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10) API Endpoints (illustrative)

Public

- GET /api/models?lang=en|ka → list with pagination/filters
- GET /api/models/:slug?lang=en|ka → detail
- GET /api/services/reverse-engineering?lang=en|ka
- GET /api/services/3d-scanning?lang=en|ka
- POST /api/contact → create submission

Admin (auth required, JWT or session)

- POST /api/auth/login
- CRUD /api/admin/models
- CRUD /api/admin/services/*
- GET /api/admin/contact-submissions
- Media upload endpoints with signed URLs

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11) UX & UI Notes

General

- Clean, light UI with emphasis on the 3D content; dark backdrop in viewer for contrast.
- Persistent language switch; store preference in cookie/localStorage.

3D Viewer

- Controls: orbit, zoom, reset camera, fullscreen; info panel (title/description).
- AR toggle near the viewer; show platform-specific hints.
- Loading states, progress bar for model downloads.

Accessibility

- All interactive controls reachable via keyboard; AR button has descriptive labels (EN/KA).

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12) Acceptance Criteria (Samples)

3D/AR Viewer

- *Given* a published model with GLB and USDZ, *when* user taps **View in AR** on iOS, *then* the model opens in AR Quick Look and returns to the page after closing.
- *Given* a published model, *when* user toggles back to 3D, *then* the inline viewer reappears with the previous camera state or default.

Localization

- *Given* the user switches from EN→KA on any page, *when* navigation occurs, *then* all UI strings and content render in KA, and language preference persists for future visits.

Admin Upload

- *Given* an admin uploads a GLB and selects **Show on → Reverse Engineering**, *when* the item is saved and published, *then* it appears on the Reverse Engineering page within 1 minute.

Contact Form

- *Given* a valid submission, *when* user clicks **Send**, *then* the record is stored, an email notification is sent to the admin inbox, and a localized success message is shown.

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13) Security Details

- Password hashing with bcrypt/argon2.
- Throttling: 5 login attempts → temporary lock; admin email alert on repeated failures.
- CSRF tokens on admin forms and contact form; CORS restricted to known origins.
- File validation: accept only whitelisted 3D/image MIME types; antivirus scan (optional) on upload.

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14) Performance Budget & Media Handling

- Enforce model size budget at upload; show warnings above 25 MB.
- Server-side or CI step to compress/optimize GLB (Draco), textures (KTX2), thumbnails (WebP/AVIF).
- Lazy-load gallery items; use responsive images (srcset).

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15) Analytics & Observability

- GA4 events: model_opened, ar_opened, contact_submitted, language_switched.
- Server logs and error tracking (Sentry or equivalent).

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16) Deployment & Environments

- **Environments**: Dev → Staging → Production.
- **CI/CD**: Lint, test, build, deploy. Rollback capability.

- **Backups**: Nightly DB and weekly full media snapshots; tested restore.

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17) Timeline (indicative)

- Week 1: IA, wireframes, visual design; set up repos, CI/CD.
- Weeks 2–3: Frontend (models gallery, detail viewer, i18n scaffolding); basic pages.
- Weeks 3–4: Admin Panel (models, services content, media library).
- Week 5: AR integrations, performance tuning, SEO, accessibility.
- Week 6: QA, UAT, content entry, launch.

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18) Open Questions / Decisions Needed

1. Branding assets (logo, color palette, typography)?
2. Do you want model downloads enabled for visitors?
3. Max file sizes for GLB/GLTF/USDZ uploads? (e.g., 50 MB hard limit)
4. Preferred cloud provider (Azure, AWS, GCP)?
5. Email service provider for contact notifications?
6. Do you need watermarking or copyright notices on images/models?
7. Should the homepage be a standalone landing or redirect to 3D Models?
8. Captcha requirement on contact form?
9. Any additional fields you want in the contact form?

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19) Appendices

A. Example UI Strings (JSON-style)

```
{
  "nav": {
    "models": {"en": "3D Models", "ka": "3D მოდელები"},
    "re": {"en": "Reverse Engineering Service", "ka": "რევერსული ინჟინერიის სერვისი"},
```

"scan": {"en": "3D Scanning Service", "ka": "3D სკანირების სერვისი"},
"contact": {"en": "Contacts", "ka": "კონტაქტი"}
},
"buttons": {
"view3d": {"en": "View in 3D", "ka": "3D ნახვა"},
"viewar": {"en": "View in AR", "ka": "AR რეჟიმი"},
"send": {"en": "Send", "ka": "გაგზავნა"}
},
"contactForm": {
"firstName": {"en": "First name", "ka": "სახელი"},
"lastName": {"en": "Last name", "ka": "გვარი"},
"email": {"en": "Email", "ka": "ელ. ფოსტა"},
"phone": {"en": "Phone", "ka": "ტელეფონი"}
},
"messages": {
"sent": {"en": "Your message has been sent. Thank you!", "ka": "თქვენი შეტყობინება გაიგზავნა. გმადლობთ!"},
"required": {"en": "Please fill in all required fields.", "ka": "გთხოვთ, შეავსოთ სავალდებულო ველები."}
}
}

B. Wireframe Notes (high level)

- **Models Gallery**: 3–4 columns desktop, 2 tablet, 1 mobile; card = thumbnail, title, tags.
- **Model Detail**: left/top = viewer (16:9); right/bottom = title, description, AR/3D toggle, metadata, images.
- **Service Pages**: hero banner + text blocks + showcases (cards or image grid).
- **Contact**: left text block with phone; right form; mobile stacks.

End of PDR